Name :- Ammar

Course :- MLT (Artificial Intelligence)

<u>Libraries :-</u>

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Statsmodels (for inference like R glm)
import statsmodels.api as sm
# sklearn (for ML evaluation)
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, roc_curve, roc_auc_score
```

<u>Sample DataFrame :-</u>

```python
df = pd.DataFrame({
    "hours": [1,2,3,4,5,6,7,8,9,10],
    "pass": [0,0,0,1,0,1,1,1,1,1]
})
```

Visualization (Logistic Curve) :-

```python
plt.scatter(df ["hours"], df["pass"])
plt.plot(df["hours"], df["pred_prob"])
plt.title("Logistic Curve: Hours vs Pass")
plt.xlabel("Hours")
plt.ylabel("Probability of Passing")
plt.show()
```

STATISTICAL Model (GLM like R) :-

```python
X = sm.add_constant(df["hours"])
y = df ["pass"]
model = sm.Logit(y, X).fit()
print (model.summary())

```

Odds Ratio :-

```
odds_ratio= np.exp(model.params)
print("\nodds Ratio:\n", odds_ratio)
```

Machine Learning Model :-

```
clf = LogisticRegression()
clf.fit(df[["hours"]], y)
```

Predictions :-

```python
df["pred_prob"] = clf.predict_proba(df[["hours"]]) [:,1]
df["pred_class"] = clf.predict(df [["hours"]])
```

Evaluation :-

```python
print("\nAccuracy:", accuracy_score (y, df["pred_class"]))
print("\nConfusion Matrix:\n", confusion_matrix(y, df ["pred_class"]))

```

ROC Curve :-

```python
fpr, tpr, _ = roc_curve (y, df["pred_prob"])
auc = roc_auc_score (y, df ["pred_prob"])

plt.plot(fpr, tpr)
plt.title(f" ROC Curve (AUC = {auc:.2f})")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.show()

```

6.BMI vs Diabetes :-

Optimization terminated successfully.

Current function value: 0.250901

Iterations 8 Logit Regression Results

==

Dep. Variable: diabetes No. Observations: 10

Model: Logit Df Residuals: 8

Method: MLE Df Model: 1

Date: Thu, 09 Apr 2026 Pseudo R-squ.: 0.6380

Time: 13:02:12 Log-Likelihood: -2.5090

converged: True LL-Null: -6.9315

Covariance Type: nonrobust LLR p-value: 0.002939

==

	coef	std err	z	P>|z|	[0.025	0.975]
const	-17.5721	11.398	-1.542	0.123	-39.912	4.767
bmi	0.6508	0.420	1.549	0.121	-0.172	1.474

==

Odds Ratio:

const 2.336289e-08

bmi 1.917111e+00

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x0000023C99014830>

[<matplotlib.lines.Line2D object at 0x0000023C99118550>]

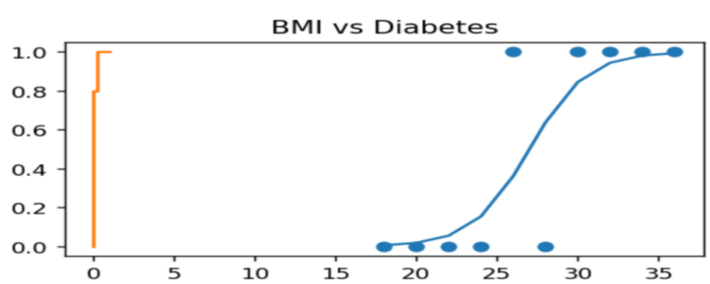
Text(0.5, 1.0, 'BMI vs Diabetes')

Accuracy: 0.8

[[4 1]

[1 4]]

[<matplotlib.lines.Line2D object at 0x0000023C9960C190>]



7. Temp vs Failiure:-

Optimization terminated successfully.

Current function value: 0.250901

Iterations 8 Logit Regression Results

==

Dep. Variable:	failure	No. Observations:	10
Model:	Logit	Df Residuals:	8
Method:	MLE	Df Model:	1
Date:	Thu, 09 Apr 2026	Pseudo R-squ.:	0.6380
Time:	13:06:35	Log-Likelihood:	-2.5090
converged:	True	LL-Null:	-6.9315
Covariance Type:	nonrobust	LLR p-value:	0.002939

==

	coef	std err	z	P>|z|	[0.025	0.975]
const	-18.8738	12.234	-1.543	0.123	-42.852	5.105
temp	0.2603	0.168	1.549	0.121	-0.069	0.590

==

Odds Ratio:

const 6.356707e-09

temp 1.297355e+00

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x0000023C996A96D0>

[<matplotlib.lines.Line2D object at 0x0000023C996A9090>]

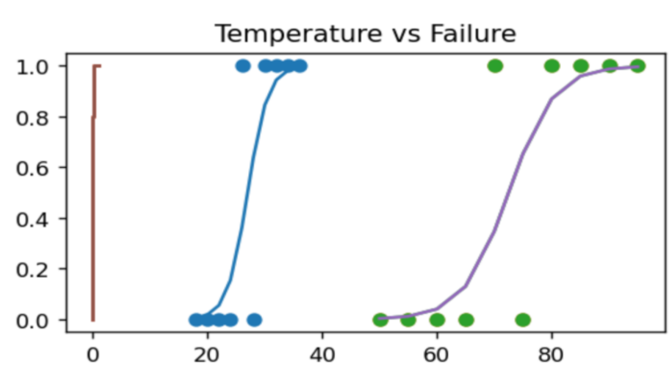
Text(0.5, 1.0, 'Temperature vs Failure')

Accuracy: 0.8

[[4 1]

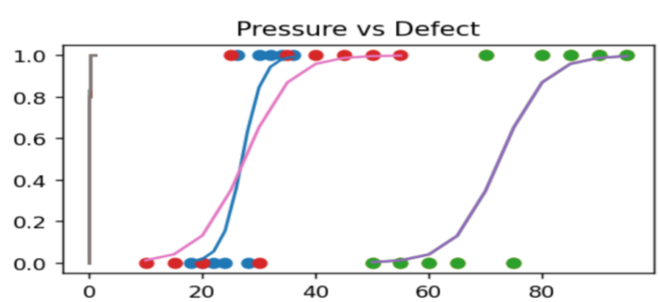
[1 4]]

[<matplotlib.lines.Line2D object at 0x0000023C996A9590>]



8. Pressure vs Defect :-

```
Optimization terminated successfully.
         Current function value: 0.250690
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:                 defect   No. Observations:                   10
Model:                          Logit   Df Residuals:                        8
Method:                           MLE   Df Model:                            1
Date:                Thu, 09 Apr 2026   Pseudo R-squ.:                  0.6275
Time:                        13:08:48   Log-Likelihood:                -2.5069
converged:                       True   LL-Null:                       -6.7301
Covariance Type:            nonrobust   LLR p-value:                  0.003658
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -7.1200      4.802     -1.483      0.138     -16.532       2.292
pressure       0.2591      0.169      1.533      0.125      -0.072       0.590
==============================================================================
Odds Ratio:
 const       0.000809
pressure    1.295747
dtype: float64
LogisticRegression()
 [<matplotlib.lines.Line2D object at 0x0000023C9960C690>]
Text(0.5, 1.0, 'Pressure vs Defect')
Accuracy: 0.8
[[3 1]
 [1 5]]
[<matplotlib.lines.Line2D object at 0x0000023C98C9F890>]
```



9. Income vs Loan :-

Optimization terminated successfully.

Current function value: 0.250690

Iterations 8 Logit Regression Results

==

Dep. Variable: loan No. Observations: 10

Model: Logit Df Residuals: 8

Method: MLE Df Model: 1

Date: Thu, 09 Apr 2026 Pseudo R-squ.: 0.6275

Time: 13:11:16 Log-Likelihood: -2.5069

converged: True LL-Null: -6.7301

Covariance Type: nonrobust LLR p-value: 0.003658

==

	coef	std err	z	P>|z|	[0.025	0.975]
const	-9.7109	6.456	-1.504	0.133	-22.365	2.943
income	0.2591	0.169	1.533	0.125	-0.072	0.590

==

Odds Ratio:

const 0.000061

income 1.295747

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x0000023C99119F90>

[<matplotlib.lines.Line2D object at 0x0000023C996A8050>]

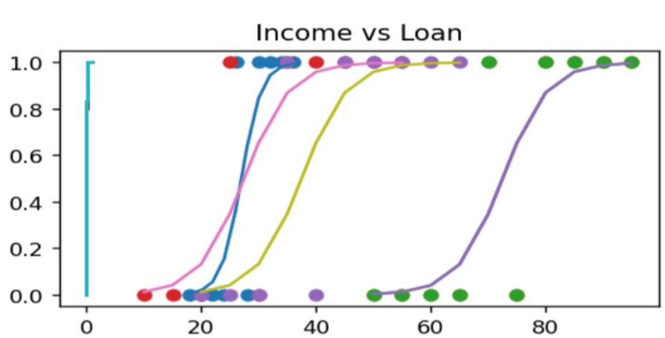
Text(0.5, 1.0, 'Income vs Loan')

Accuracy: 0.8

[[3 1]

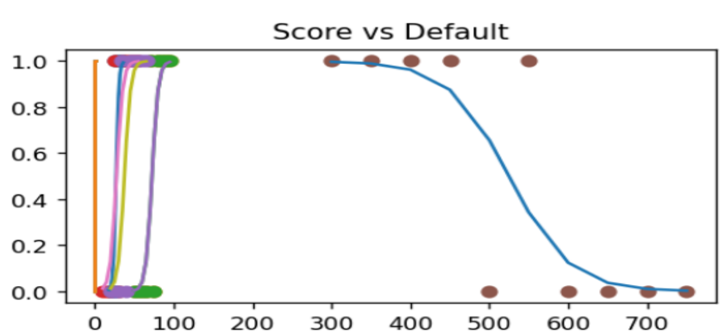
[1 5]]

[<matplotlib.lines.Line2D object at 0x0000023C996A8550>]]



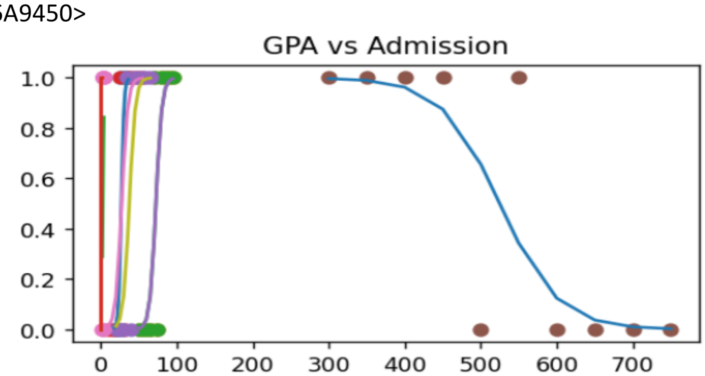
10. Credit Score vs Default:-

```
Optimization terminated successfully.
         Current function value: 0.250901
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:                default   No. Observations:                   10
Model:                          Logit   Df Residuals:                        8
Method:                           MLE   Df Model:                            1
Date:                Thu, 09 Apr 2026   Pseudo R-squ.:                  0.6380
Time:                        13:13:48   Log-Likelihood:                -2.5090
converged:                       True   LL-Null:                       -6.9315
Covariance Type:            nonrobust   LLR p-value:                  0.002939
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         13.6672      8.894      1.537      0.124      -3.765      31.099
score         -0.0260      0.017     -1.549      0.121      -0.059       0.007
==============================================================================
Odds Ratio:
 const    862165.225392
score         0.974303
dtype: float64
LogisticRegression()
<matplotlib.collections.PathCollection object at 0x0000023C996A9E50>
[<matplotlib.lines.Line2D object at 0x0000023C996A9950>]
Text(0.5, 1.0, 'Score vs Default')
Accuracy: 0.8
[[4 1]
 [1 4]]
[<matplotlib.lines.Line2D object at 0x0000023C996A9BD0>]
```



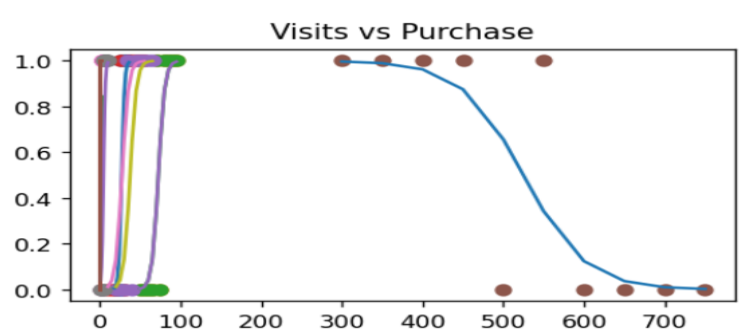
11. GPA vs Admission :-

```
Optimization terminated successfully.
         Current function value: 0.270909
         Iterations 8                    Logit Regression Results
==============================================================================
Dep. Variable:                  admit   No. Observations:                   10
Model:                          Logit   Df Residuals:                        8
Method:                           MLE   Df Model:                            1
Date:                Thu, 09 Apr 2026   Pseudo R-squ.:                  0.5975
Time:                        13:16:18   Log-Likelihood:                -2.7091
converged:                       True   LL-Null:                       -6.7301
Covariance Type:            nonrobust   LLR p-value:                  0.004570
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const        -13.8057      8.653     -1.595      0.111     -30.766       3.155
gpa            4.8499      2.998      1.618      0.106      -1.026      10.726
==============================================================================
Odds Ratio:
 const      0.000001
gpa      127.730367
dtype: float64
LogisticRegression()
<matplotlib.collections.PathCollection object at 0x0000023C996A9450>
[<matplotlib.lines.Line2D object at 0x0000023C996A9D10>]
Text(0.5, 1.0, 'GPA vs Admission')
Accuracy: 0.9
[[3 1]
 [0 6]]
[<matplotlib.lines.Line2D object at 0x0000023C996A9F90>]
```



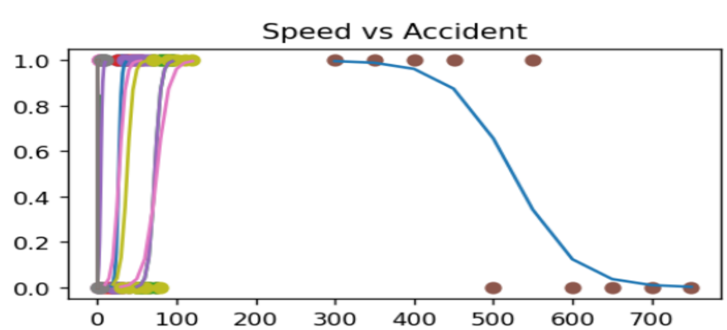
12. Visit vs Purchase:-

```
Optimization terminated successfully.
         Current function value: 0.250690
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:                    buy   No. Observations:                   10
Model:                          Logit   Df Residuals:                        8
Method:                           MLE   Df Model:                            1
Date:                Thu, 09 Apr 2026   Pseudo R-squ.:                  0.6275
Time:                        13:18:04   Log-Likelihood:                -2.5069
converged:                       True   LL-Null:                       -6.7301
Covariance Type:            nonrobust   LLR p-value:                  0.003658
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -5.8246      3.986     -1.461      0.144     -13.637       1.988
visits         1.2954      0.845      1.533      0.125      -0.361       2.952
==============================================================================
Odds Ratio:
 const     0.002954
visits    3.652592
dtype: float64
LogisticRegression()
<matplotlib.collections.PathCollection object at 0x0000023C996AA0D0>
[<matplotlib.lines.Line2D object at 0x0000023C996AA210>]
Text(0.5, 1.0, 'Visits vs Purchase')
Accuracy: 0.8
[[3 1]
 [1 5]]
[<matplotlib.lines.Line2D object at 0x0000023C996A9810>]
```



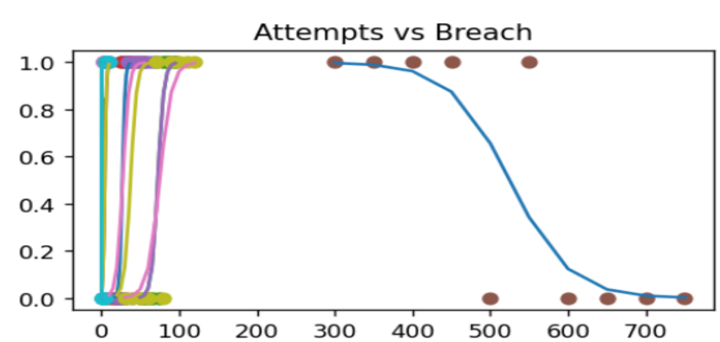
13. Speed vs Accident :-

```
Optimization terminated successfully.
         Current function value: 0.250901
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:             accident   No. Observations:                   10
Model:                        Logit   Df Residuals:                        8
Method:                         MLE   Df Model:                            1
Date:              Thu, 09 Apr 2026   Pseudo R-squ.:                  0.6380
Time:                      13:19:37   Log-Likelihood:                -2.5090
converged:                     True   LL-Null:                       -6.9315
Covariance Type:          nonrobust   LLR p-value:                  0.002939
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -9.7623      6.403     -1.525      0.127     -22.312       2.787
speed          0.1302      0.084      1.549      0.121      -0.034       0.295
==============================================================================
Odds Ratio:
 const    0.000058
speed    1.139015
dtype: float64
LogisticRegression()
<matplotlib.collections.PathCollection object at 0x0000023C996AA490>
[<matplotlib.lines.Line2D object at 0x0000023C996AA350>]
Text(0.5, 1.0, 'Speed vs Accident')
Accuracy: 0.8
[[4 1]
 [1 4]]
[<matplotlib.lines.Line2D object at 0x0000023C996AA5D0>]
```



14. Attempt vs Breach :-

```
Optimization terminated successfully.

        Current function value: 0.250690

        Iterations 8                    Logit Regression Results

==============================================================================

Dep. Variable:              breach   No. Observations:               10

Model:                       Logit   Df Residuals:                    8

Method:                        MLE   Df Model:                        1

Date:             Thu, 09 Apr 2026   Pseudo R-squ.:              0.6275

Time:                     13:21:11   Log-Likelihood:            -2.5069

converged:                    True   LL-Null:                   -6.7301

Covariance Type:         nonrobust   LLR p-value:              0.003658

==============================================================================

               coef    std err          z      P>|z|      [0.025      0.975]

------------------------------------------------------------------------------

const       -5.8246      3.986     -1.461      0.144     -13.637       1.988

attempts     1.2954      0.845      1.533      0.125      -0.361       2.952

==============================================================================

Odds Ratio:

 const       0.002954

attempts    3.652592

dtype: float64

LogisticRegression()

<matplotlib.collections.PathCollection object at 0x0000023C996AA710>

[<matplotlib.lines.Line2D object at 0x0000023C996AA990>]

Text(0.5, 1.0, 'Attempts vs Breach')

Accuracy: 0.8

[[3 1]

 [1 5]]

[<matplotlib.lines.Line2D object at 0x0000023C996AAC10>]
```



15. Exercise vs Weight Loss :-

```
Optimization terminated successfully.
         Current function value: 0.250690
         Iterations 8
                           Logit Regression Results
==============================================================================
Dep. Variable:                   loss   No. Observations:                   10
Model:                          Logit   Df Residuals:                        8
Method:                           MLE   Df Model:                            1
Date:                Thu, 09 Apr 2026   Pseudo R-squ.:                  0.6275
Time:                        13:22:49   Log-Likelihood:                -2.5069
converged:                       True   LL-Null:                       -6.7301
Covariance Type:            nonrobust   LLR p-value:                  0.003658
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -5.8246      3.986     -1.461      0.144     -13.637       1.988
hours          1.2954      0.845      1.533      0.125      -0.361       2.952
==============================================================================
Odds Ratio:
 const    0.002954
hours    3.652592
dtype: float64
LogisticRegression()
<matplotlib.collections.PathCollection object at 0x0000023C996AAAD0>
[<matplotlib.lines.Line2D object at 0x0000023C996AAE90>]
Text(0.5, 1.0, 'Exercise vs Weight Loss')
Accuracy: 0.8
[[3 1]
 [1 5]]
[<matplotlib.lines.Line2D object at 0x0000023C996AAD50>]
```